

ASSIGNMENT-01

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Branch and Section: CSE-A

Subject: Computer networks.

I. Short Answer Questions

1] Define network and Computer network.

A. Network: A network is a collection of Interconnected device that can communicate and share resources.

Computer network: A Computer network is a group of computers and other devices connected through communication links to exchange data and share resource such as files, printers, and the internet.

2. Mention Different types of Transmission media.

A. Transmission media are of two types.

1. Guided (wired) media.

- Twisted pair cable
- Coaxial cable
- Optical Fiber cable

2. Unguided (wireless) media

- Radio waves
- microwave
- Infrared
- satellite Communication

3. Define the internet?

A. The internet is a worldwide network of Interconnected computer networks that communicate using the TCP/IP protocol to share information and services.

4. State framing?
A. Framing is a data link layer process of dividing a stream of bits into identifiable units called frames by adding headers and trailers for reliable transmission.

5. List out the error detection methods.

A. The common error detection methods are

- parity check (VRC),
- Longitudinal Redundancy Check (LRC)
- Checksum
- Cyclic Redundancy Check (CRC)

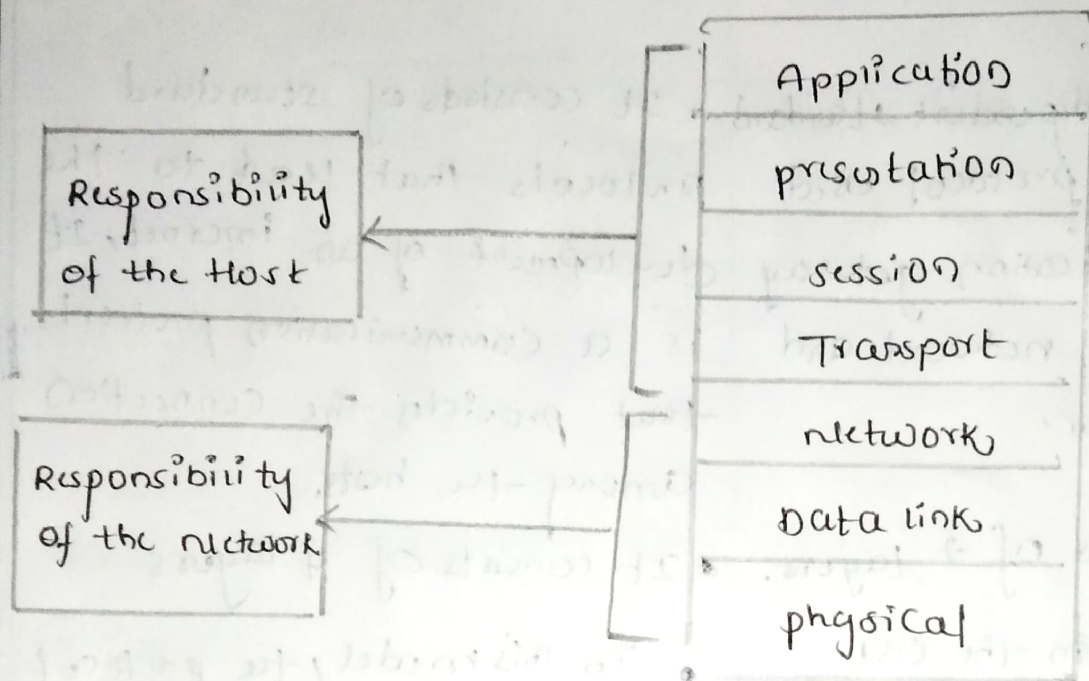
II Long Answer Questions

1. Explain about OSI Reference model with neat diagram.

A. OSI Reference model:-

- OSI stands for Open System Interconnection is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.
- consists of seven layers, and each layer performs a particular network function.

- OSI model was developed by the international organization for standardization (ISO) in 1984
- OSI model divides the whole task into seven smaller and manageable tasks.



- The OSI model is divided into two layers.
 - upper layers and
 - lower layers.
 - 1) upper layers implemented only in the software related
 - 2) lower layer of the OSI model deals with the data transport issues.
 - The data link layer and the physical layer are implemented in hardware and software
2. Compare and contrast OSI and Tcp/ip reference model

A.

OSI model

- It stands for open system Inter connection.
- OSI model has been developed by OSI.
- It is an independent standard and generic protocol used as a Communication gateway between the network and the end users.
- It consists of 7 layers.
- protocols in the OSI models are hidden and can be easily replaced when the technology changes.

TCP/IP model

- It stands for Transmission Control protocol.
- It was developed by ARPANET.
- It consists of standard protocols that leads to the development of an internet, It is a communication protocol that provides the connection among two hosts.
- It consists of 4 layers.
- In this model, the protocol cannot be easily replaced.

3. Suppose we want to transmit the message 110010010001 and protect it from error using the CRC polynomial $x^3 + x^2 + 1$ long division to determine the message that should be transmitted.

A) Given,

110010010001

polynomial: $x^3 + x^2 + 1 = 1101$

$$\begin{array}{r}
 1101 \overline{) 1100100101001000} \\
 \underline{1101} \\
 0001100 \\
 \underline{1101} \\
 0001101 \\
 \underline{1101} \\
 0000001000 \\
 \underline{1101} \\
 01010 \text{ Final remainder}
 \end{array}$$

$$\begin{array}{r}
 \rightarrow 1010 \\
 \underline{1001} \\
 0011
 \end{array}$$

Bring down the next bit

$$\begin{array}{r}
 \rightarrow 1100 \\
 \underline{1101} \\
 0001 \rightarrow \text{bring down the last bit} = 0010
 \end{array}$$

no more bits remains

original state = 1100100101001

CRC = 001

Transmitted message = 1100100101001001

4. Explain Error Detecting Techniques with Example

A. Error Detection: Error detection is the process of identifying whether the transmitted data has been changed or corrupted during transmission.

Types of error detecting methods

1) parity check (VRC - Vertical Redundancy Check):

- A parity check bit is added to the data to make the total number of 1's either even or odd.

Ex:

data = 1011001

number of 1's = 4 (even)

parity bit = 0

Transmitted data = 10110010

Final answer = 10110010

Number of 1's become 5 (odd), hence an error is detected

2) Checksum:

The sender divides the data into equal-sized blocks, adds them using binary addition, and blocks, and sends the 1's complement of the sum as the checksum.

Data blocks

1010

1100

addition

1010

1100

10110

checksum = 01001

The receiver adds all blocks including the checksum

sum = All 1's $\rightarrow$ no error

Otherwise $\Rightarrow$ error detected

5. Cyclic Redundancy Check (CRC): CRC uses binary polynomial division to detect transmission errors.

Ex:

Data = 11100000

Generator = 1001

Append 3 zeros

1110000000

modulo-2 Division

CRC Remainder = 001

Transmitted data = 11100000001

At the Receiver? If the remainder is 000, the data is correct.

• Otherwise, an error is detected.

5. If the even parity Hamming Code (0100010111000) is transmitted and the received code is (010001011000), determine whether the received code is correct or not. justify

A: Given,

Transmitted (T) = 0100010111000

Received code (R) = 0100010110000

parity = even parity

T = 0 1 0 0 0 1 0 1 1 1 0 0 0

R = 0 1 0 0 0 1 0 1 1 0 0 0 0

↑
Error at 10th bit

The transmitted and received code differ at the 10th bit

Check parity bit:

In hamming code, parity bits are located at position

1, 2, 4, 8.

The receiver recalculates these parity bits

= 1010

$$1010_2 = 10_{10}$$

The syndrome value indicates that the 10th bit contains an error

Flip the 10th bit.

Received Code: 010001011000

Corrected code: 01000111000

Final Answer:

$$\text{Syndrome} = 1010_2 = 10$$

Error position = 10th bit

Corrected code = 010001011000